DEVIXO SOLUTIONS

Web Development Internship Program

TASK 01

Professional Personal Portfolio Website

Student Name	Hassan Asad
Internship Domain	Web Development
Task Number	Task 01

1. Objective

The objective of this task was to design and develop a modern, professional, and fully responsive personal portfolio website using HTML5 and CSS3. The website had to showcase my personal profile, technical skills, education, projects, and contact information through a clean and attractive web interface.

The task required the website to be built from scratch with my own HTML and CSS. Bootstrap, Tailwind CSS, website templates, and other UI frameworks were not allowed, so every layout, component, and animation on the site was written by hand.

By completing this task I aimed to practice the fundamentals of professional web development:

HTML5, CSS3, Layouts, Responsive Design, Navigation, Animations, UI Design, and GitHub.

2. Technologies Used

Technology	How it was used
HTML5	Page structure using semantic elements (header, nav, main, section, footer), forms with labels, and descriptive alt text on images.
CSS3	All styling in a single style.css file: Flexbox and CSS Grid layouts, CSS custom properties (variables), media queries, transitions, keyframe animations, and the hover effects.
Git and GitHub	Version control and hosting of the final source code, images, screenshots, resume, and documentation.
Visual Studio Code	Code editor used to write and format the HTML and CSS.
Browser developer tools	Responsive mode used to test the layout at desktop, laptop, tablet, and mobile widths.

No JavaScript, CSS frameworks, UI libraries, or downloaded templates were used in the website.

3. Website Structure

3.1 Folder structure

The project files are organized as follows:

```
portfolio/
|-- index.html
|-- style.css
|-- README.md
|-- images/
|   |-- logo.svg
|   |-- profile.png
|   |-- project1.png
|   |-- project2.jpg
|   |-- project3.png
|   |-- icons/   (language, tool, and social icons)
|-- resume/
|   |-- Hassan_Asad_Resume.pdf
|-- screenshots/
|   |-- (desktop and mobile screenshots)
|-- report/
    |-- Task01_Report.pdf
```

File or folder	Purpose
index.html	The complete page structure and all content.
style.css	All styling, layout, responsive rules, and animations.
images/	Profile photo, logo, and project thumbnails.
images/icons/	Icons for the languages marquee and the GitHub and LinkedIn cards.
resume/	PDF resume linked from the hero section.
screenshots/	Screenshots used in the README and this report.
README.md	Project documentation shown on the GitHub repository page.

3.2 Page structure

The website is a single page. A sticky navigation bar sits inside a header element, the six sections are placed inside main, and a footer closes the page. Every section has an id so the navigation links can jump straight to it.

Section	Contents
Navigation bar	Logo on the left and links to Home, About, Skills, Education, Projects, and Contact.
Home	Name, professional tagline, short introduction, call to action button (My Projects), resume button, profile photo with a rotating ring, and a terminal style info box.
About	A large motivational statement on the left. On the right: personal introduction, academic background, career interests, and a brief professional summary.
Skills	Six skill cards grouped by category, followed by a scrolling marquee showing the languages I know with their icons.

Section	Contents
Education	Cards for my BS in Software Engineering, Intermediate in Computer Science, and two training courses, each with the institute, study period, and a description.
Projects	Three project cards, each with an image, short description, technologies used, and a GitHub button.
Contact	Email, phone, and location, large GitHub and LinkedIn cards, and a contact form with Name, Email, Subject, Message, and a Submit button.
Footer	Copyright, my name, quick navigation links, and social links.

4. Design Approach

4.1 Color theme

I used a dark theme with one purple accent color. Using only five colors keeps the design consistent and professional, and the white text on very dark backgrounds gives strong contrast.

Role	Color	Used for
Background	#060606	Main page background and the navigation bar
Accent	#8C52FF	Buttons, highlights, active borders, and glow effects
Surface	#242326	Cards, the contact form, and the terminal box
Muted gray	#5E5F5F	Card borders and small details
Text	#FFFFFF	Headings and body text (softened white for secondary text)

4.2 Typography and visual hierarchy

The site uses a clean system font stack (Segoe UI, with fallbacks to system fonts) so text stays readable on every device. A clear size difference between the hero name, section headings, card titles, and body text guides the reader through the page. Section headings have an underline, and accent colored labels separate categories inside cards.

4.3 Layout and spacing

- CSS Grid is used for the hero section, skill cards, education cards, project cards, about section, and contact layout, so cards line up and wrap automatically.
- Flexbox is used for the navigation bar, tags, buttons, footer, and aligning content inside cards.
- Spacing, border radius, and transition speed are stored in CSS variables so every section uses the same values.
- Sections share the same maximum width and padding, which keeps the page aligned from top to bottom.

4.4 Responsive design

The layout adapts to desktop, laptop, tablet, and mobile screens. Grids use flexible column sizes, images scale within their containers, and two media query breakpoints adjust the layout. The page does not create unnecessary horizontal scrolling.

Device	Width	Layout behavior
Desktop	1200px and above	Two column hero and about sections, three project columns, two column contact layout, full navigation bar.
Laptop	901px to 1199px	Same structure as desktop with cards wrapping into fewer columns as space gets smaller.
Tablet	up to 900px	Hero becomes a single column with the photo on top, the about and contact sections stack, and cards wrap into one or two columns.
Mobile	up to 600px	Reduced section padding, centered navigation, single column cards, and smaller headings and form padding. On touch screens the project extra information is always visible because hover is not available.

4.5 Hover animations

Simple, smooth CSS transitions give feedback without being excessive. Users who prefer reduced motion get a version with animations turned off.

Element	Effect
Navigation links	Text brightens and an accent underline appears.
Buttons (call to action, resume, GitHub, Submit)	Button lifts slightly and changes color, with a soft glow on the main buttons.
Skill cards	Card moves up, border turns to the accent color, and a glow appears.
Education cards	Card lifts slightly with a soft glow.
Project cards	Card lifts and scales up with a stronger glow, and a panel with extra project information fades in over the image.
Social cards	Card lifts and the border changes to the accent color.
Profile photo	Dashed ring rotates slowly and the terminal cursor blinks.

4.6 Accessibility and usability

- Every image has descriptive alt text and every form field has a label connected to it.
- Visible focus outlines are provided for keyboard users.
- The navigation has an aria-label and links use clear, meaningful text.
- The prefers-reduced-motion setting turns off transitions and animations.

5. Website Features

Navigation and general

- Sticky navigation bar that stays visible while scrolling, with a logo and working links to every section.
- Smooth scrolling between sections, with offset so headings are not hidden behind the navigation bar.
- Consistent dark color theme and spacing throughout the page.

Home section

- Name, professional tagline, and a short introduction.
- Call to action button that takes visitors to the projects, and a resume button that opens my resume PDF.
- Profile photo inside a slowly rotating dashed ring with a small orbiting dot.
- Terminal style info box showing my name, role, stack, and degree.

About section

- Large statement on the left about being motivated and quick to learn.
- Personal introduction, academic background, career interests, and professional summary on the right.

Skills section

- Six skill cards: programming languages, backend development, frontend development, databases, AI and machine learning, and tools and core concepts.
- A moving marquee of the languages I know, each with its own icon.

Education section

- Four large cards showing the degree or course, institute, study period, relevant details, and a short description.

Projects section

- Three consistent project cards: SmartCourse, TaskFlow, and Personal Portfolio.
- Each card has a project image, short description, technologies used, and a GitHub button.
- Hovering over a card makes it pop up and reveals extra information about the project.

Contact section

- Email, phone, and location details.
- Large GitHub and LinkedIn cards with icons for quick access to my profiles.

- Contact form UI with Name, Email, Subject, Message, and Submit button. The form does not need backend functionality.

Footer

- Copyright notice, my name, quick navigation links, and social links.

6. Code Screenshots

The following screenshots show part of the HTML and CSS source code.

HTML source code

index.html | style.css | README.md

index.html > html > body > main > section#contact > div.contact-layout > div.contact-left > div.social-links

```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <meta name="description" content="Hassan Asad, Software Engineering Graduate and Python/Django Backend Developer">
    <title>Hassan Asad | Developer Portfolio</title>
    <link rel="icon" type="image/svg+xml" href="images\logo.svg">
</head>
<link rel="stylesheet" href="style.css">
</head>

<body>

    <!-- navbar -->
    <header>
        <nav id="navbar" aria-label="Main navigation">
            <div class="navbar-inner">
                <a class="nav-logo" href="#home">
                    <span class="nav-logo-icon">
                        <img src="images\logo.svg" alt="H logo">
                    </span>
                </a>

                <ul id="navbar-list">
                    <li class="navbar-list-item"><a href="#home">Home</a></li>
                    <li class="navbar-list-item"><a href="#about">About</a></li>
                    <li class="navbar-list-item"><a href="#skills">Skills</a></li>
                    <li class="navbar-list-item"><a href="#education">Education</a></li>
                    <li class="navbar-list-item"><a href="#projects">Projects</a></li>
                    <li class="navbar-list-item"><a href="#contact">Contact</a></li>
                </ul>
            </div>
        </nav>
    </header>

    <main>
```

Figure 1: Navbar in index.html

index.html | style.css | README.md

index.html > html > body > main > section#contact > div.contact-layout > div.contact-left > div.social-links

```
<html lang="en">
<body>
    <main>

        <!-- skills -->
        <section id="skills">
            <h2 id="skills-heading">Skills</h2>

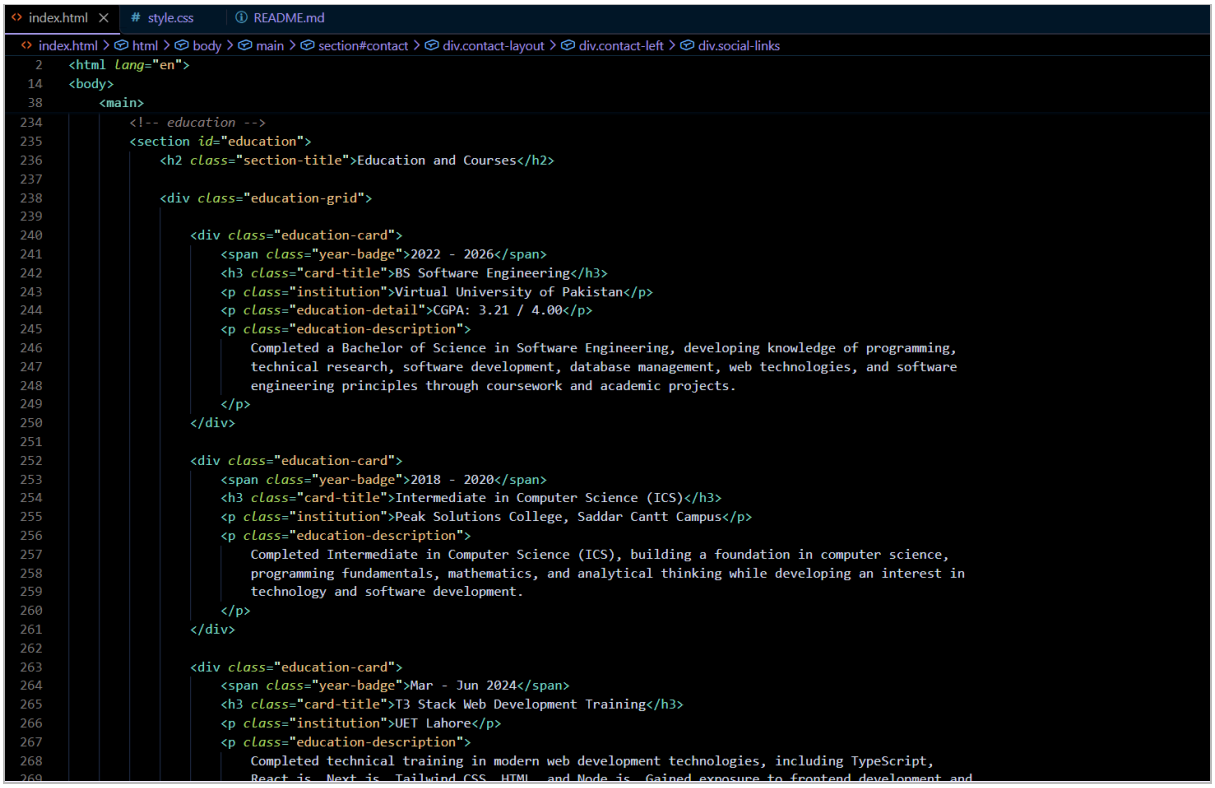
            <div class="skills-grid">

                <div class="skill-card">
                    <h3 class="skill-card-heading">Programming Languages</h3>
                    <div class="skills-list">
                        <span class="skill-tag">Python</span>
                        <span class="skill-tag">JavaScript</span>
                        <span class="skill-tag">TypeScript</span>
                        <span class="skill-tag">C++</span>
                        <span class="skill-tag">PHP</span>
                    </div>
                </div>

                <div class="skill-card">
                    <h3 class="skill-card-heading">Backend Development</h3>
                    <div class="skills-list">
                        <span class="skill-tag">Flask</span>
                        <span class="skill-tag">Django</span>
                        <span class="skill-tag">Django REST Framework</span>
                        <span class="skill-tag">REST APIs</span>
                    </div>
                </div>

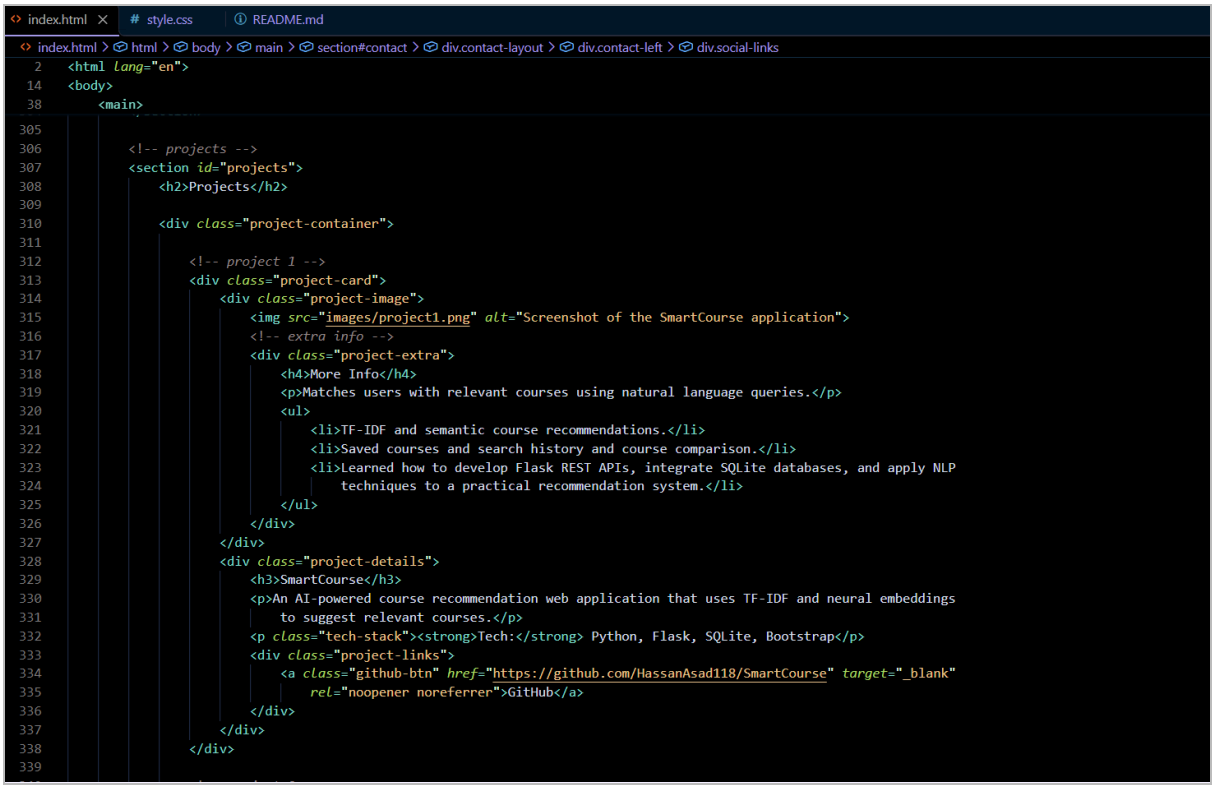
                <div class="skill-card">
                    <h3 class="skill-card-heading">Frontend Development</h3>
                    <div class="skills-list">
                        <span class="skill-tag">HTML and CSS</span>
                        <span class="skill-tag">Bootstrap</span>
                        <span class="skill-tag">React.js</span>
                        <span class="skill-tag">Next.js</span>
```

Figure 2: Skills Section code in index.html



```
index.html  style.css  README.md
index.html > html > body > main > section#contact > div.contact-layout > div.contact-left > div.social-links
  2  <html lang="en">
 14  <body>
 38      <main>
234          <!-- education -->
235          <section id="education">
236              <h2 class="section-title">Education and Courses</h2>
237
238              <div class="education-grid">
239
240                  <div class="education-card">
241                      <span class="year-badge">2022 - 2026</span>
242                      <h3 class="card-title">BS Software Engineering</h3>
243                      <p class="institution">Virtual University of Pakistan</p>
244                      <p class="education-detail">CGPA: 3.21 / 4.00</p>
245                      <p class="education-description">
246                          Completed a Bachelor of Science in Software Engineering, developing knowledge of programming,
247                          technical research, software development, database management, web technologies, and software
248                          engineering principles through coursework and academic projects.
249                      </p>
250                  </div>
251
252                  <div class="education-card">
253                      <span class="year-badge">2018 - 2020</span>
254                      <h3 class="card-title">Intermediate in Computer Science (ICS)</h3>
255                      <p class="institution">Peak Solutions College, Saddar Cantt Campus</p>
256                      <p class="education-description">
257                          Completed Intermediate in Computer Science (ICS), building a foundation in computer science,
258                          programming fundamentals, mathematics, and analytical thinking while developing an interest in
259                          technology and software development.
260                      </p>
261                  </div>
262
263                  <div class="education-card">
264                      <span class="year-badge">Mar - Jun 2024</span>
265                      <h3 class="card-title">T3 Stack Web Development Training</h3>
266                      <p class="institution">UET Lahore</p>
267                      <p class="education-description">
268                          Completed technical training in modern web development technologies, including TypeScript,
269                          React.js, Next.js, Tailwind CSS, HTML, and Node.js. Gained exposure to frontend development and
```

Figure 3: Education section in index.html



```
index.html  style.css  README.md
index.html > html > body > main > section#contact > div.contact-layout > div.contact-left > div.social-links
  2  <html lang="en">
 14  <body>
 38      <main>
305
306          <!-- projects -->
307          <section id="projects">
308              <h2>Projects</h2>
309
310              <div class="project-container">
311
312                  <!-- project 1 -->
313                  <div class="project-card">
314                      <div class="project-image">
315                          <img src="images/project1.png" alt="Screenshot of the SmartCourse application">
316                          <!-- extra info -->
317                          <div class="project-extra">
318                              <h4>More Info</h4>
319                              <p>Matches users with relevant courses using natural language queries.</p>
320                              <ul>
321                                  <li>TF-IDF and semantic course recommendations.</li>
322                                  <li>Saved courses and search history and course comparison.</li>
323                                  <li>Learned how to develop Flask REST APIs, integrate SQLite databases, and apply NLP
324                                      techniques to a practical recommendation system.</li>
325                              </ul>
326                          </div>
327                      </div>
328                      <div class="project-details">
329                          <h3>SmartCourse</h3>
330                          <p>An AI-powered course recommendation web application that uses TF-IDF and neural embeddings
331                              to suggest relevant courses.</p>
332                          <p class="tech-stack"><strong>Tech:</strong> Python, Flask, SQLite, Bootstrap</p>
333                          <div class="project-links">
334                              <a class="github-btn" href="https://github.com/HassanAsad118/SmartCourse" target="_blank"
335                                  rel="noopener noreferrer">GitHub</a>
336                          </div>
337                      </div>
338                  </div>
339
```

Figure 4: Projects section in index.html

CSS source code

index.html | style.css | README.md

style.css > ...

```css
/* colors and fonts */
:root {
    --bg: #060606;
    --accent: #8C52FF;
    --surface: #242326;
    --muted: #5E5F5F;
    --text: #FFFFFF;

    --accent-light: #A67DFF;
    --text-soft: rgba(255, 255, 255, 0.75);

    --font-main: "Segoe UI", system-ui, -apple-system, Roboto, Helvetica, Arial, sans-serif;

    --radius: 10px;
    --max-width: 1200px;
    --transition: 0.25s ease;
}

/* reset and base */
*,
*::before,
*::after {
    box-sizing: border-box;
    margin: 0;
    padding: 0;
}

html {
    scroll-behavior: smooth;
    scroll-padding-top: 70px;
}

body {
    font-family: var(--font-main);
    background-color: var(--bg);
    color: var(--text);
    line-height: 1.7;
    overflow-x: hidden;
}
```

Figure 3: CSS variables and base styles in style.css

index.html M | style.css | README.md

style.css > ...

```css
.education-description {
    margin: 12px 0 16px;
}

/* projects */
.project-container {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(min(100%, 340px), 1fr));
    gap: 32px;
}

.project-card {
    display: flex;
    flex-direction: column;
    background-color: var(--surface);
    border: 1px solid var(--muted);
    border-radius: var(--radius);
    overflow: hidden;
    transition: transform 0.35s ease, box-shadow 0.35s ease, border-color 0.35s ease;
}

/* the pop up effect */
.project-card:hover,
.project-card:focus-within {
    transform: translateY(-12px) scale(1.03);
    border-color: var(--accent);
    box-shadow: 0 22px 44px rgba(140, 82, 255, 0.3);
    position: relative;
    z-index: 2;
}

.project-image {
    position: relative;
    height: 260px;
    overflow: hidden;
    background-color: var(--bg);
    border-bottom: 1px solid var(--muted);
}
```

Figure 4: Project Cards in style.css

index.html M | # style.css | README.md

style.css > ...

```
.footer-socials a:hover {
}

/* tablet */
@media (max-width: 900px) {
    #home {
        grid-template-columns: 1fr;
        min-height: auto;
        text-align: center;
    }

    #my-name,
    #tagline,
    #intro,
    .hero-buttons,
    .hero-visual {
        grid-column: 1;
    }

    #intro {
        margin-left: auto;
        margin-right: auto;
    }

    .about-layout {
        grid-template-columns: 1fr;
        gap: 36px;
    }

    .hero-buttons {
        justify-content: center;
    }

    /* photo and terminal go on top on smaller screens */
    .hero-visual {
        grid-row: 1;
        margin-bottom: 36px;
    }
```

Figure 5: Responsive media queries in style.css

index.html M | # style.css | README.md

style.css > ...

```
/* mobile */
@media (max-width: 600px) {
    section {
        padding: 56px 16px;
    }

    section h2,
    .section-title {
        font-size: 1.6rem;
    }

/* ************************** */

    .navbar-inner {
        justify-content: center;
        padding: 10px 16px;
    }

    #navbar-list {
        justify-content: center;
        gap: 6px 16px;
    }

/* ********************** */

    .navbar-list-item a {
        font-size: 0.85rem;
    }

    .about-right {
        padding-left: 20px;
    }

    .education-card {
        padding: 24px;
        min-height: auto;
    }
```

Figure 6: Responsive media queries in style.css

7. Desktop Screenshots

These screenshots show every section of the website on a desktop screen.

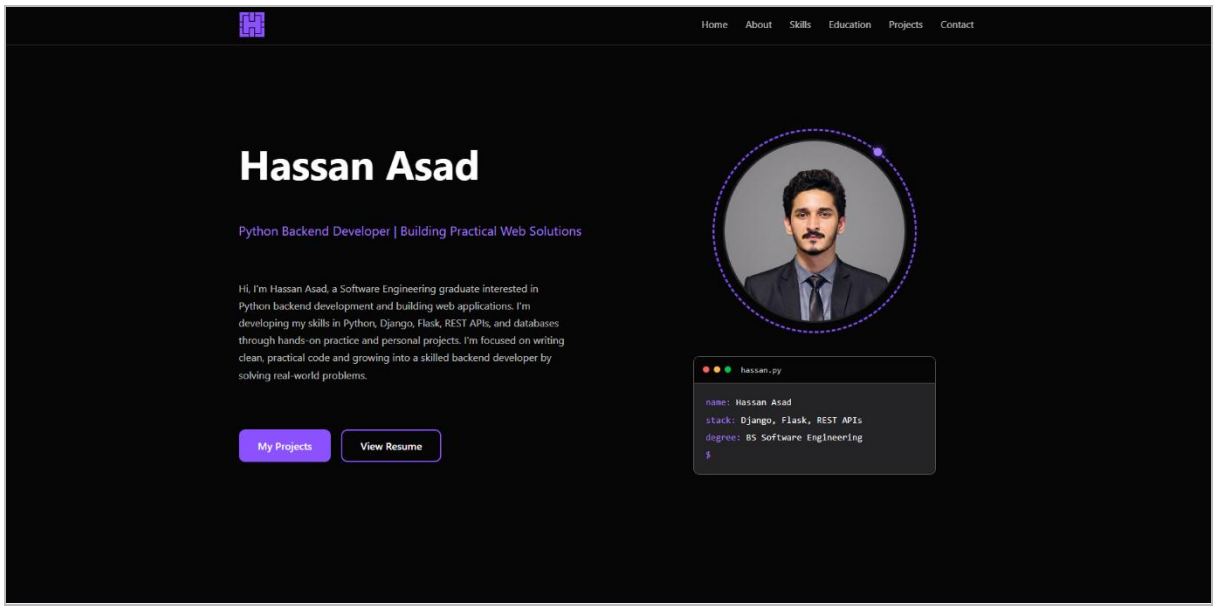


Figure 5: Home hero section

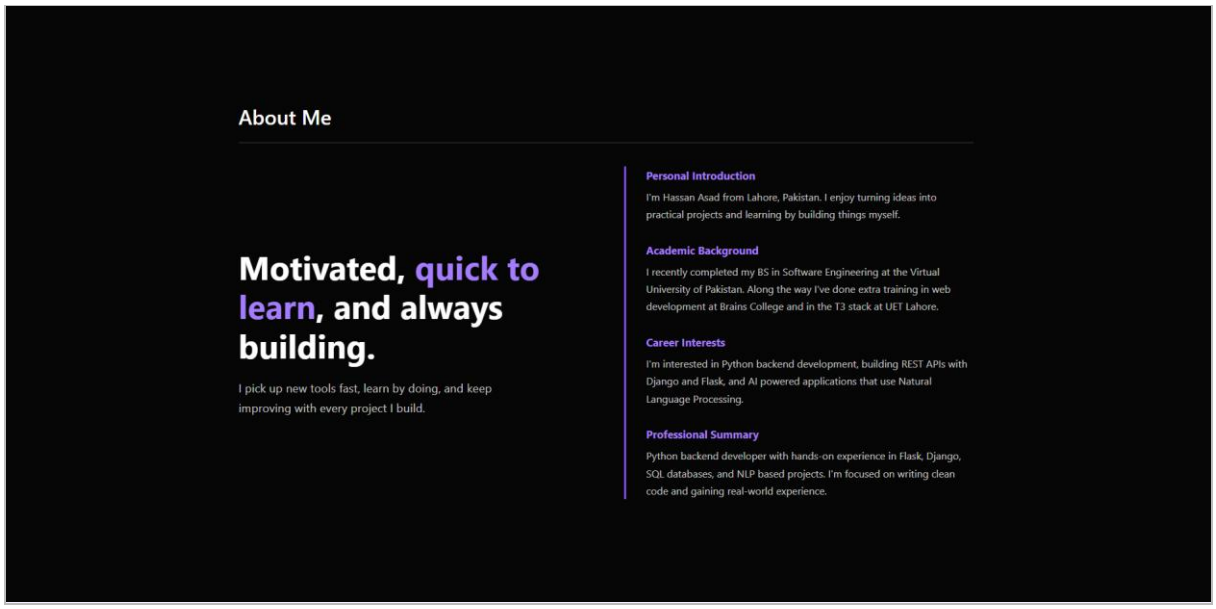


Figure 6: About section

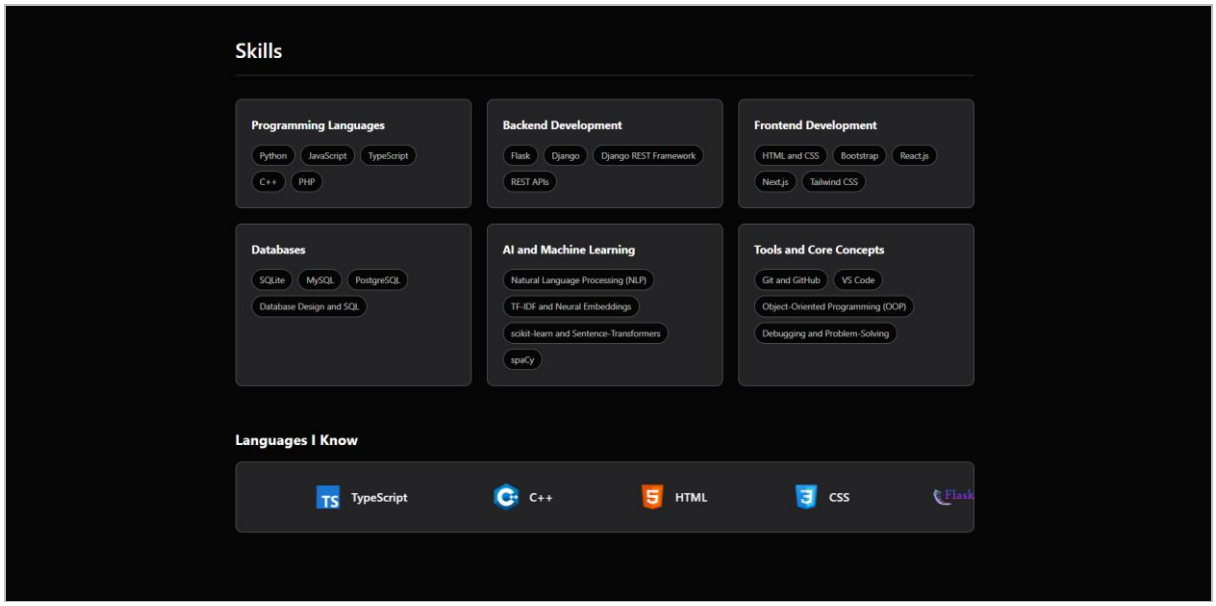


Figure 7: Skills section

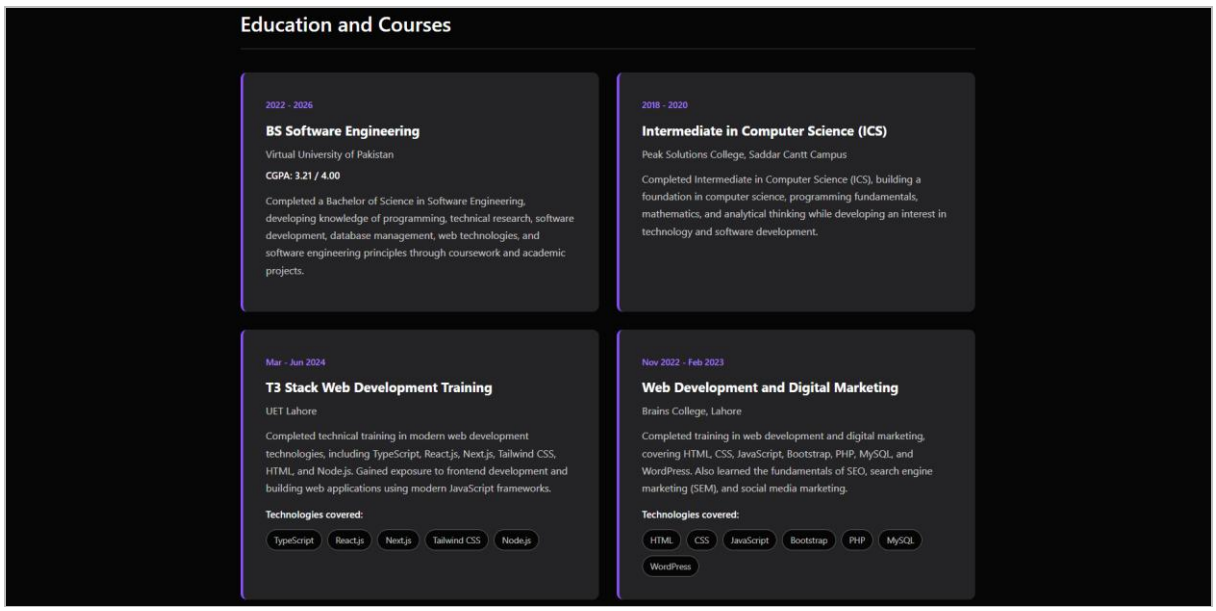


Figure 8: Education section

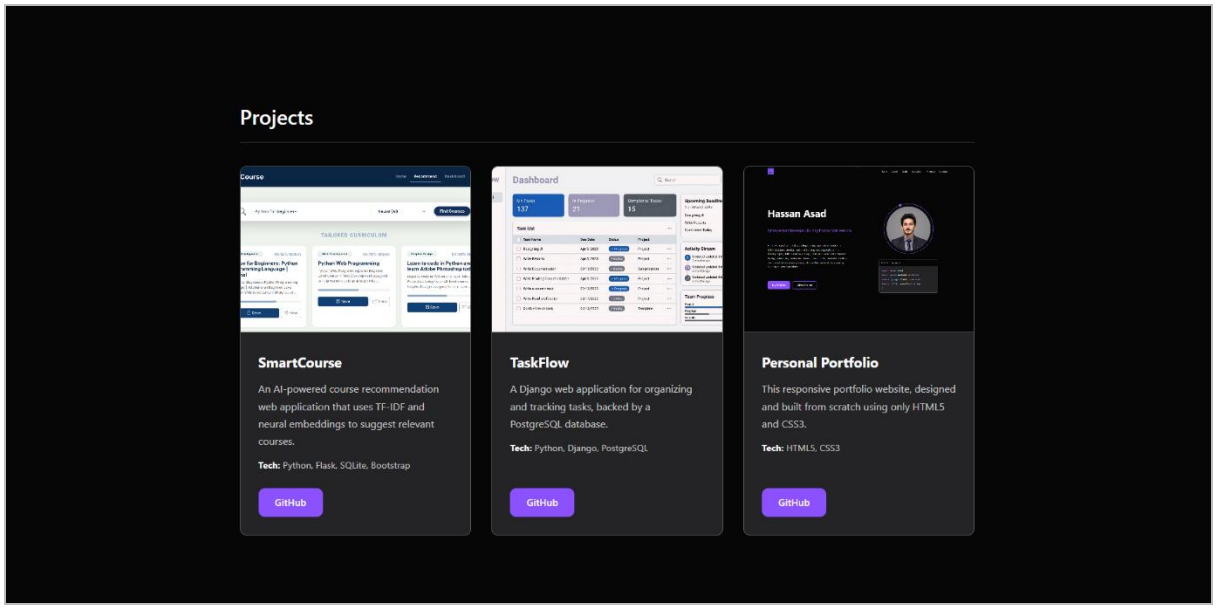


Figure 9: Projects section

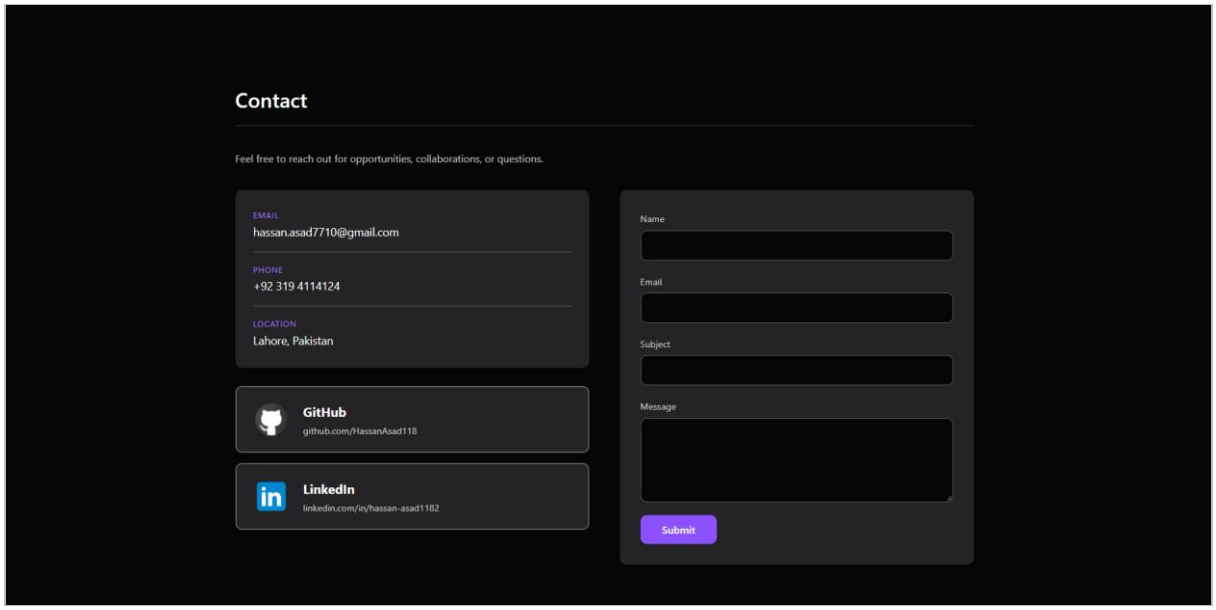


Figure 10: Contact section

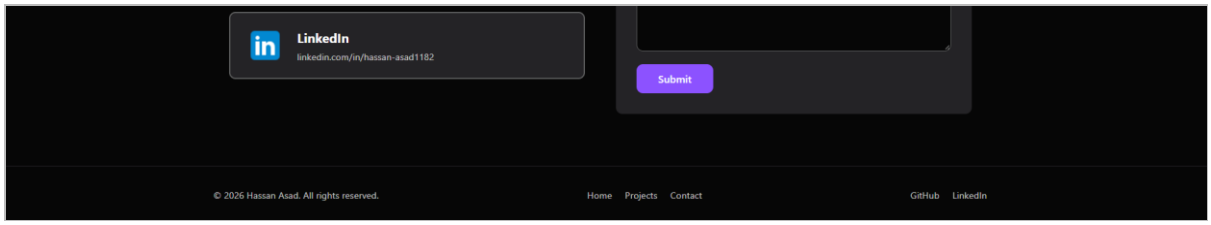


Figure 11: Footer

8. Mobile Screenshots

The website was checked at a mobile width of 375px. The layout switches to a single column and no horizontal scrolling appears.

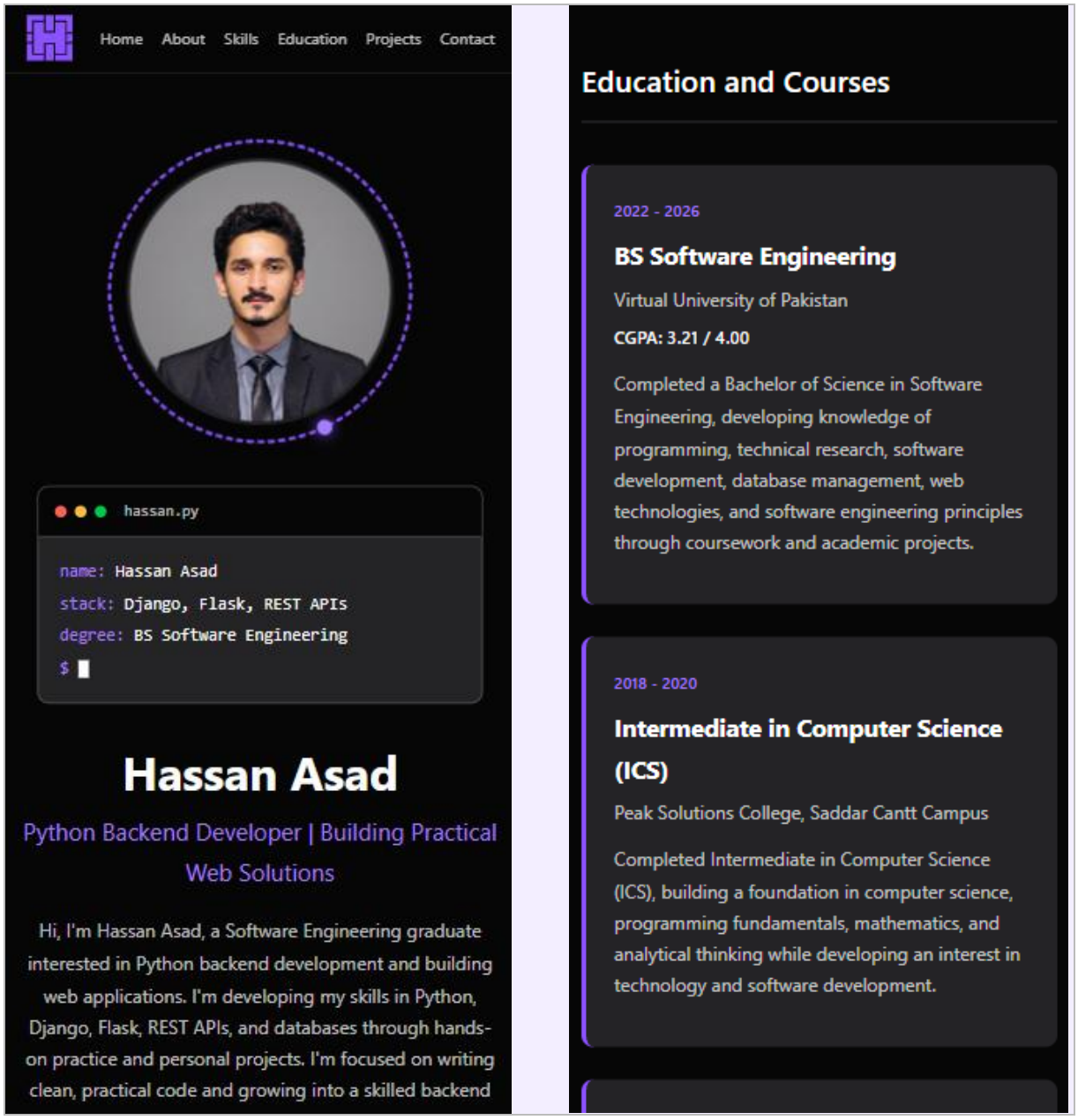


Figure 12: Website on a mobile screen

9. Conclusion

This task gave me the chance to build a complete, professional portfolio website from scratch using only HTML5 and CSS3. I practiced structuring a page with semantic HTML, creating layouts with CSS Grid and Flexbox, making a design fully responsive with media queries, and adding smooth hover effects and animations that improve the user experience without being excessive.

I also learned the importance of a consistent color theme, clear visual hierarchy, clean code organization, and meaningful naming. Using Git and GitHub to organize and publish the project, along with writing documentation, taught me how a real web project is delivered.

The final website presents my skills, education, and projects clearly and professionally, and it gives me a strong foundation that I can keep extending as I build new projects.

10. GitHub Repository Link

Repository: https://github.com/HassanAsad118/Portfolio.git

The repository contains the HTML source code, CSS source code, images and assets, this PDF report, screenshots, and the README.md file.